MILITARY STANDARD

MIL-STD-1325-39A

RAILCAR LOADING OF HAZARDOUS MATERIALS

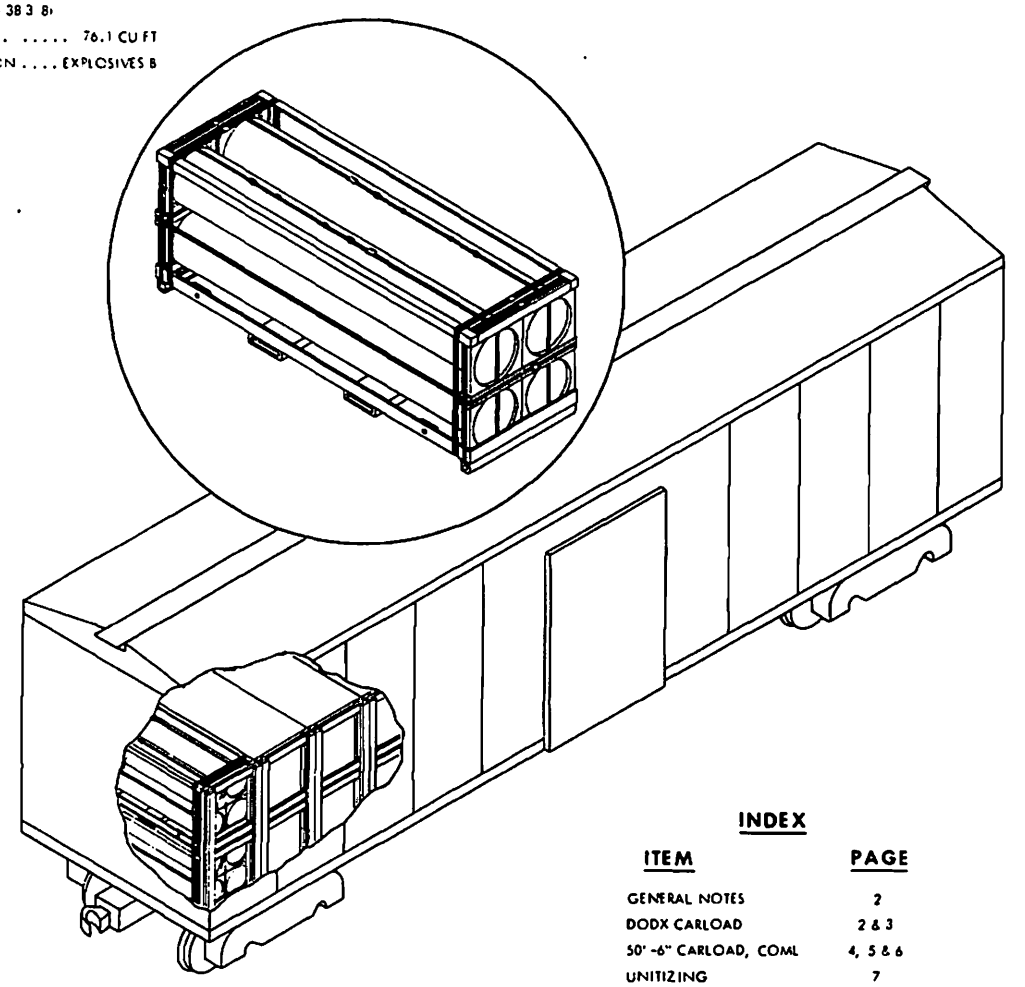
ROCKET MOTOR CLUSTER LAU-10/A, LAU-10A/A,
LAU-10B/A OR LAU-10C/A LAUNCHER (ZUNI)
IN UNIT LOAD ADAPTER MK58 MOD 1
FLEET ISSUE UNIT LOAD

7 JUNE 1974

SUPERSEDING
MIL-STD-1325-39
(NAVY)
27 MARCH 1973

UNIT LOAD DATA

UNIT LOAD DRAWING WR-54 115
WEIGHT ..... SEE TABLE I
DIMENSIONS ..... 99-1 2 L x 3-: W x 38-7 8 H
(STACKING HEIGHT - 38 3 8)
CUBE ..... 76.1 CU FT
DOT HAZARD CLASSIFICATION .... EXPLOSIVES B



INDEX

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GENERAL NOTES	2
DODX CARLOAD	2 & 3
50'-6" CARLOAD, COML	4, 5 & 6
UNITIZING	7

NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCK-LOADING AND CONTAINERLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS MIL-HDBK-236 (NAVY).

FSC 8140

AUTHORIZED AND RELEASED FOR GENERAL USE.

W. F. Grassmuck — MILITARY ASSISTANT — 4/3/74
SIGNATURE — DATE

W. S. Gay NIOHL
SIGNATURE — TECHNICAL DIRECTION AGENT (TDA) — DATE

E. McC.
SIGNATURE — AIRSYSCOM BY DIRECTION — DATE

APPROVED BY BUREAU OF EXPLOSIVES

ORIGINATOR — Charles Gallo 4/10/74
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
N A D EARLE, NEW JERSEY

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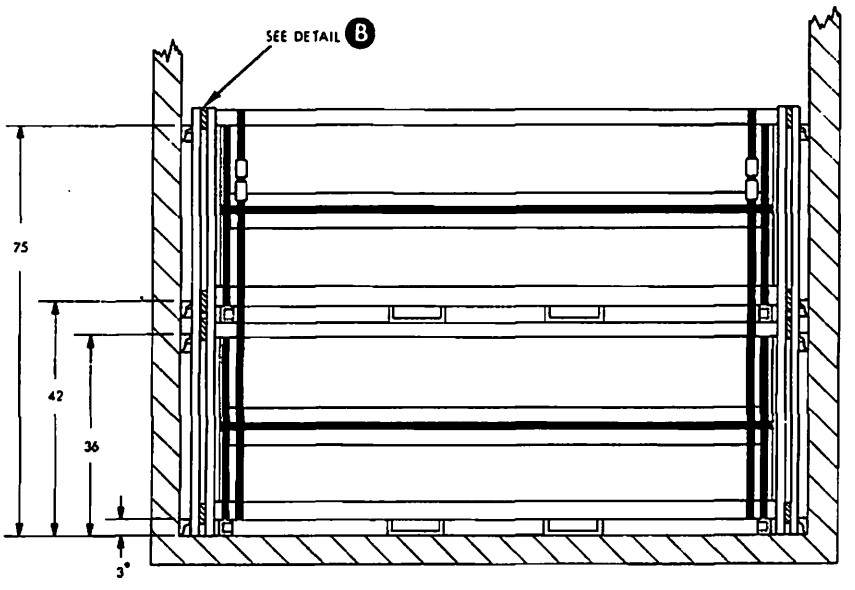
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GENERAL NOTES

1. FOR GENERAL INFORMATION CONCERNING ORDERING, INSPECTING, AND PREPARING CARS, AND FOR DUNNAGING MATERIALS, DESIGN, AND INSTALLATION OF DUNNAGE SEE THE GENERAL DOCUMENT MIL-STD-1325 (NAVY) "RAILCAR LOADING OF HAZARDOUS MATERIALS".
2. WHEN PLANNING SHIPMENTS ORDER THE MINIMUM NUMBER OF CARS OF THE CAPACITY REQUIRED FOR THE SHIPMENT. UTILITY LOADER CARS SHALL BE SERIES DODX 28000.
3. LOADING PLANS SHOWN ARE FOR DODX UTILITY LOADER CAR WITH 50 FT. 6 INCHES INSIDE LENGTH, 107-3/4 INCHES INSIDE WIDTH BETWEEN RAILS (111 INCHES INSIDE WIDTH BETWEEN SIDE WALLS), AND COMMERCIAL BOXCARS WITH 50 FT. 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH.
4. THE "LOAD LIMIT" OF A CAR MUST NOT BE EXCEEDED NOR SHOULD THE RAILCAR BE LOADED SO THAT MORE THAN ONE-HALF OF THE "LOAD LIMIT" IS CARRIED BY ONE SET OF TRUCKS.
5. IF END WALLS OF CARS ARE NOT SQUARE THEY MUST BE SQUARED OFF BEFORE STARTING TO LOAD CAR.
6. THE LOAD CONSISTS OF LAU-10/A, LAU-10A/A OR LAU-10B/A LAUNCHERS PALLETIZED IN ACCORDANCE WITH WR-54/115
7. THE UNIT LOADS ARE HANDLED AND LOADED WITH A SUITABLE FORK LIFT TRUCK.
8. UNLESS OTHERWISE SPECIFIED NAILING SHALL BE IN ACCORDANCE WITH MIL-STD-1325 (NAVY).
9. ALL STRAP JOINTS SHALL BE CRIMP-TYPE JOINTS MADE WITH A CRIMPING TOOL THAT DOES NOT CUT THE SEAL OR STRAP, RATHER THAN A TOOL THAT MAKES A NOTCH-TYPE JOINT. THE STRAPPING MUST BE IN ACCORDANCE WITH NAVSUP NOTICE 4642 DATED 21 APRIL 1971.
10. APPLICABLE MATERIAL SPECIFICATIONS:
    DUNNAGE LUMBER - FED. SPEC MM-L-751
    NAILS - FED. SPEC QQ-S-761, TYPE I, HEAVY DUTY, CLASS A, DRY (UNLUBRICATED).
    STRAPPING - FED. SPEC QQ-S-781, TYPE I, HEAVY DUTY, CLASS A, DRY (UNLUBRICATED).
    SEALS - FED. SPEC QQ-S-761, STYLE III, HEAVY DUTY
11. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS INSIDE THE CAR IN AN ACCESSIBLE AREA, CLOSE AND SEAL BOXCAR DOORS, AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF CAR AS PRESCRIBED IN OP 2165 (VOL I).

50 FT 6 IN. BOXCAR DODX

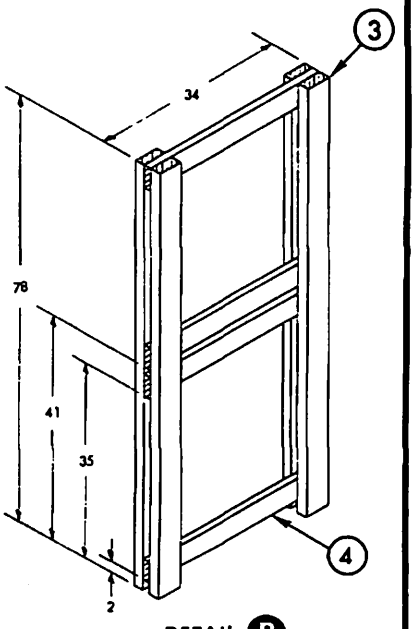
1. THE CARLOAD CONSISTS OF 32 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. A DETAILED DESCRIPTION AND OPERATING INSTRUCTIONS FOR THE UTILITY LOADER ARE CONTAINED IN OP 1750
3. UNIT LOADS SHALL BE STACKED TWO HIGH AND STRAPPED TOGETHER AS DESCRIBED ON PAGE 7.
4. TO PREVENT UNUSED "DF" EQUIPMENT FROM BECOMING DISLODGED DURING TRANSIT OF DODX CARS, SECURE IT AT ANY LOCATION IN THE BOXCAR WHICH WILL NOT INTERFERE WITH UNLOADING.
5. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN DODX COXCARS THE SAME PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE. ANY BAYS OR PORTION THEREOF MAY BE USED PROVIDING THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR ARE COMPLIED WITH. SEE MIL-STD-1325 (NAVY). EACH CROSSMEMBER WILL BE USED IN SUCH A MANNER THAT IT WILL RETAIN NOT MORE THAN 2000 LBS. OF THE LADING.



*5" HEIGHT IN DOORWAY AREA

SECTION A-A

SHOWING LOCATION OF WALL MEMBERS USED FOR CROSSMEMBERS

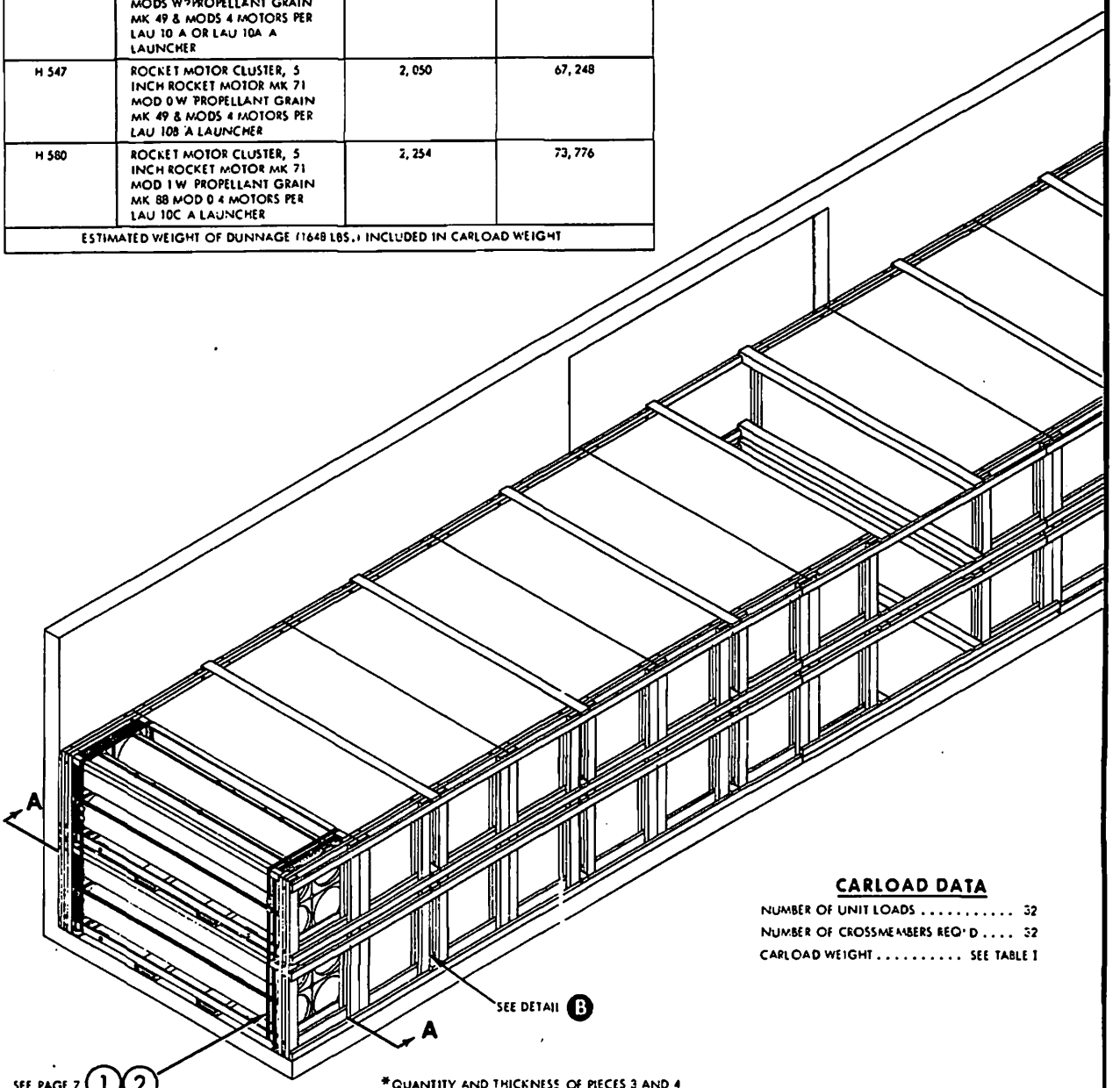


DETAIL B

SWAY BRACE FRAME
32 REQUIRED

TABLE I

DODIC NALC	ITEM DESCRIPTION	UNIT LOAD WT. (LBS., APPROX.)	CARLOAD WT. (LBS., APPROX.)
H 546	ROCKET MOTOR CLUSTER, 5 INCH ROCKET MOTOR MK 16 & MODS W?PROPELLANT GRAIN MK 49 & MODS 4 MOTORS PER LAU 10 A OR LAU 10A A LAUNCHER	2,014	66,096
H 547	ROCKET MOTOR CLUSTER, 5 INCH ROCKET MOTOR MK 71 MOD 0 W PROPELLANT GRAIN MK 49 & MODS 4 MOTORS PER LAU 108 A LAUNCHER	2,050	67,248
H 580	ROCKET MOTOR CLUSTER, 5 INCH ROCKET MOTOR MK 71 MOD 1 W PROPELLANT GRAIN MK 88 MOD 0 4 MOTORS PER LAU 10C A LAUNCHER	2,254	73,776
ESTIMATED WEIGHT OF DUNNAGE (1648 LBS.) INCLUDED IN CARLOAD WEIGHT			



CARLOAD DATA

NUMBER OF UNIT LOADS 32
NUMBER OF CROSSMEMBERS REQ'D 32
CARLOAD WEIGHT ..... SEE TABLE I

*QUANTITY AND THICKNESS OF PIECES 3 AND 4 MAY HAVE TO BE VARIED TO SUIT WIDTH OF BOXCAR. (DO NOT EXCEED 1/4" SPACE EACH SIDE OF CAR.)

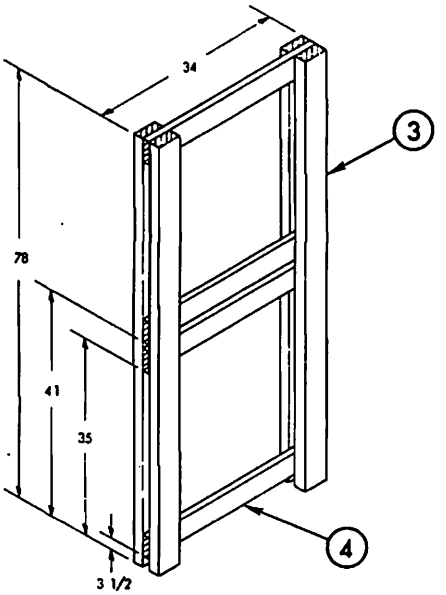
PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQ'D	NAIL TO	NAILS NUMBER	NAILS SIZE
4*	SWAY BRACE HORIZONTAL	1 x 4 x 34	128	SEE 3	-	-
3*	SWAY BRACE VERTICAL	2 x 4 x 78	128	4	3 PER JOINT	10d
2	SEAL	1 1/4	64	-	-	-
1	STRAP	1 1/4 x .035 x 19 FT. 6	32	-	-	-

LIST OF MATERIALS AND NAILING DATA

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50 FT 6 IN. BOXCAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF 34 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52 100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED. THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR MUST BE COMPLIED WITH, SEE MIL-STD-1325 (NAVY).
3. UNIT LOADS SHALL BE STACKED TWO HIGH AND STRAPPED TOGETHER AS DESCRIBED ON PAGE 7.
4. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT. WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT. WIDE PROVIDING THE LENGTH OF PIECE 9 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING.
5. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG TYPE DOORS PROVIDING THE LENGTH OF PIECE 9 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING. DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR, WHETHER MAIN OR AUXILIARY, EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR, AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR. IF LUMBER OF SUFFICIENT LENGTH TO SPAN PLUG DOORS IS NOT AVAILABLE, RANDOM LENGTH MATERIAL, DOUBLED AND SPLICED, BUT WITH JOINTS OF SPLICES OFFSET, MAY BE USED. DOORWAY PROTECTION PIECES 11 AND 12, IS NOT REQUIRED WHEN PLUG DOOR EQUIPPED BOXCARS ARE USED, EXCEPT WHEN CAR HAS A COMBINATION OF PLUG DOOR AND CONVENTIONAL SLIDING DOOR. THEN DOORWAY PROTECTION IS REQUIRED FOR THE CONVENTIONAL DOOR. SECURELY CLOSE DOORS AND WIRE TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.



DETAIL A

SWAY BRACE FRAME

34 REQUIRED

*QUANTITY AND THICKNESS OF PIECES 3 AND 4 MAY HAVE TO BE VARIED TO SUIT WIDTH OF BOXCAR. (DO NOT EXCEED 1.4" SPACE EACH SIDE OF CAR.)

**2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4'S

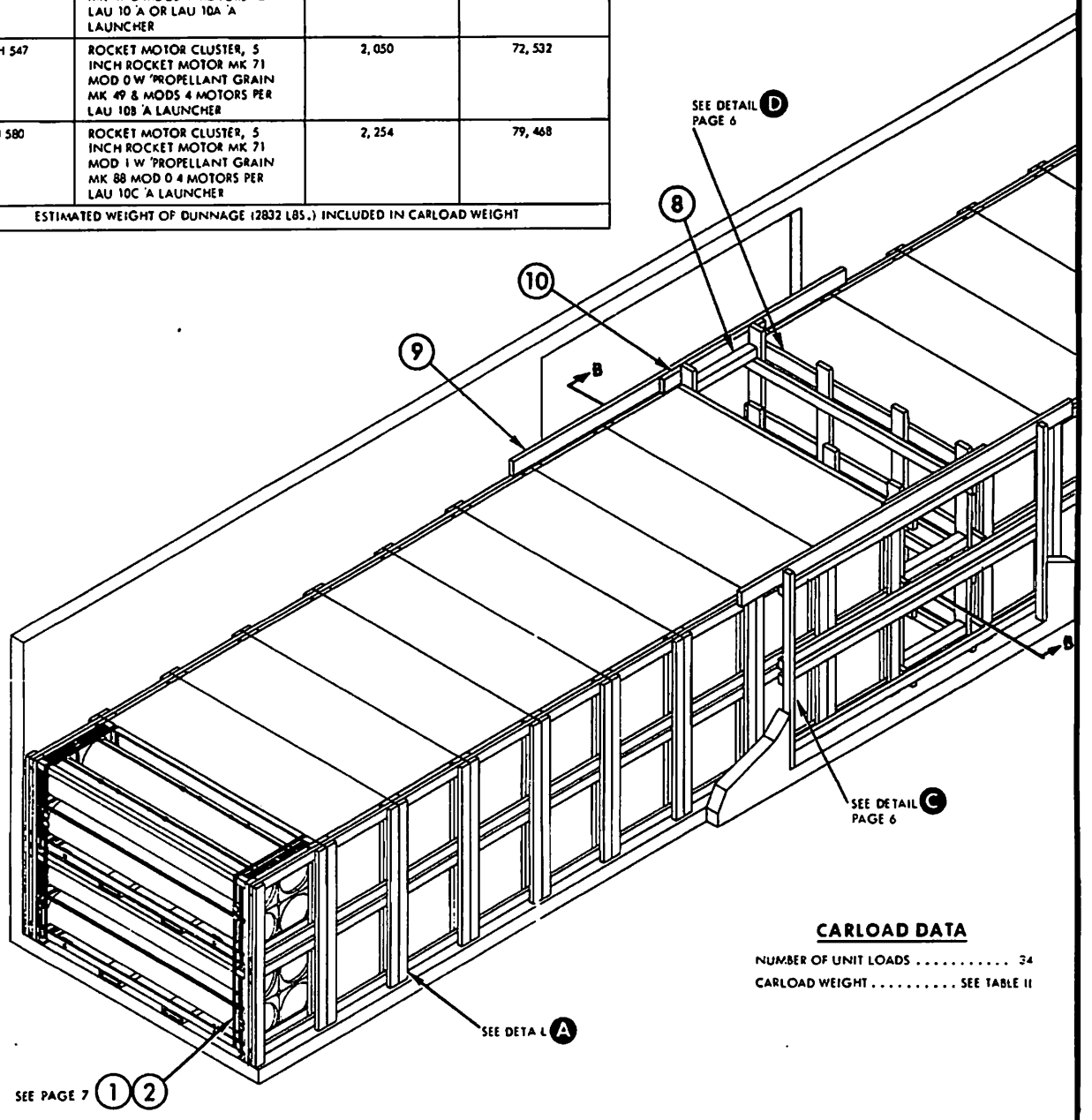
PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQ'D	NAIL TO	NAILS NUMBER	NAILS SIZE
12	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	8	11	2 PER JOINT	16d
11	VERTICAL DOORWAY MEMBER	2 x 3 x 84	4	DOOR POST	2 PER FOOT	20d
10	HOLD DOWN CLEAT	2 x 6 x 48	2	9	2 PER FOOT	10d
9	GATE HOLD DOWN	2 x 6 x 14 FT.	2	CAR WALL	5 EACH END	10d
8	STRUT	4 x 4 x WEDGE FIT**	12	6	2 PER JOINT	16d
7	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH - 1	8	6	3 PER JOINT	10d
6	CENTER GATE VERTICAL	2 x 6 x 84	8	SEE 5	-	-
5	CENTER GATE HORIZONTAL	2 x 5 x CAR WIDTH - 1	8	6	3 PER JOINT	10d
4*	SWAY BRACE HORIZONTAL	2 x 4 x 24	136	SEE 3	-	-
3*	SWAY BRACE VERTICAL	2 x 4 x 78	136	4	3 PER JOINT	10d
2	SEAL	1 1/4	68	-	-	-
1	STRAP	1 1/4 x .035 x 19 FT. 6	34	-	-	-

LIST OF MATERIALS AND NAILING DATA

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TABLE II

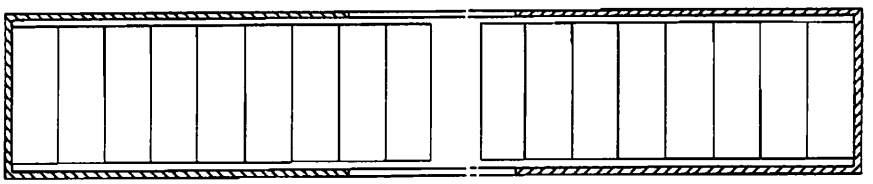
DODIC NALC	ITEM DESCRIPTION	UNIT LOAD WT. (LBS., APPROX.)	CARLOAD WT. (LBS., APPROX.)
H 546	ROCKET MOTOR CLUSTER, 5 INCH ROCKET MOTOR MK 16 & MODS W 'PROPELLANT GRAIN MK 49 & MODS 4 MOTORS PER LAU 10 'A OR LAU 10A 'A LAUNCHER	2,014	71,308
H 547	ROCKET MOTOR CLUSTER, 5 INCH ROCKET MOTOR MK 71 MOD 0 W 'PROPELLANT GRAIN MK 49 & MODS 4 MOTORS PER LAU 10B 'A LAUNCHER	2,050	72,532
H 580	ROCKET MOTOR CLUSTER, 5 INCH ROCKET MOTOR MK 71 MOD 1 W 'PROPELLANT GRAIN MK 88 MOD 0 4 MOTORS PER LAU 10C 'A LAUNCHER	2,254	79,468
ESTIMATED WEIGHT OF DUNNAGE (2832 LBS.) INCLUDED IN CARLOAD WEIGHT			



CARLOAD DATA

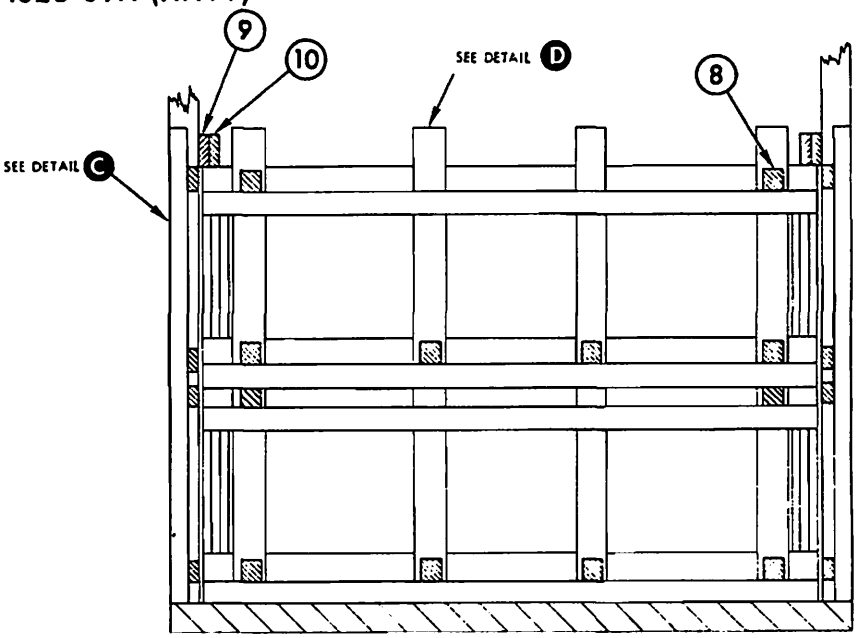
NUMBER OF UNIT LOADS 34

CARLOAD WEIGHT . . . . . SEE TABLE II



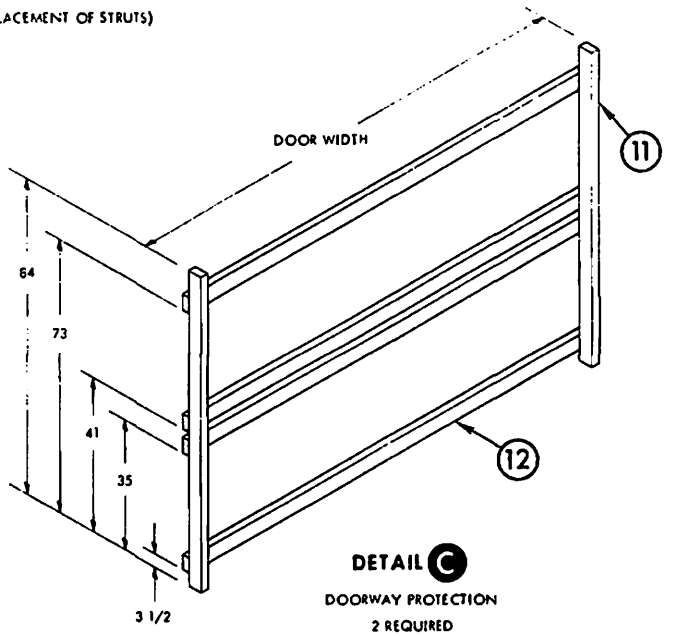
LOADING PLAN

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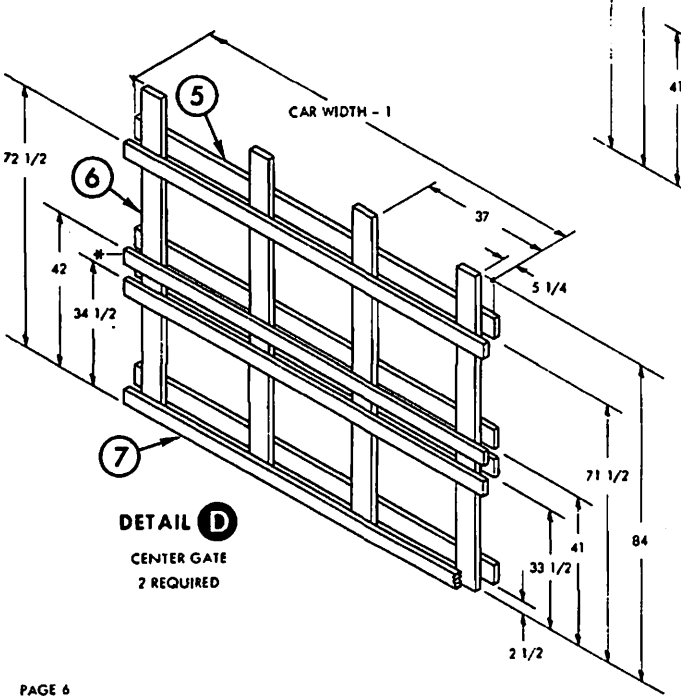
SECTION B-B
(NOTE PLACEMENT OF STRUTS)

*POSITION AFTER STRUTS HAVE BEEN INSTALLED ON LOWER STRUT CLEAT (REQUIRED ONE GATE ONLY)



DETAIL C
DOORWAY PROTECTION
2 REQUIRED

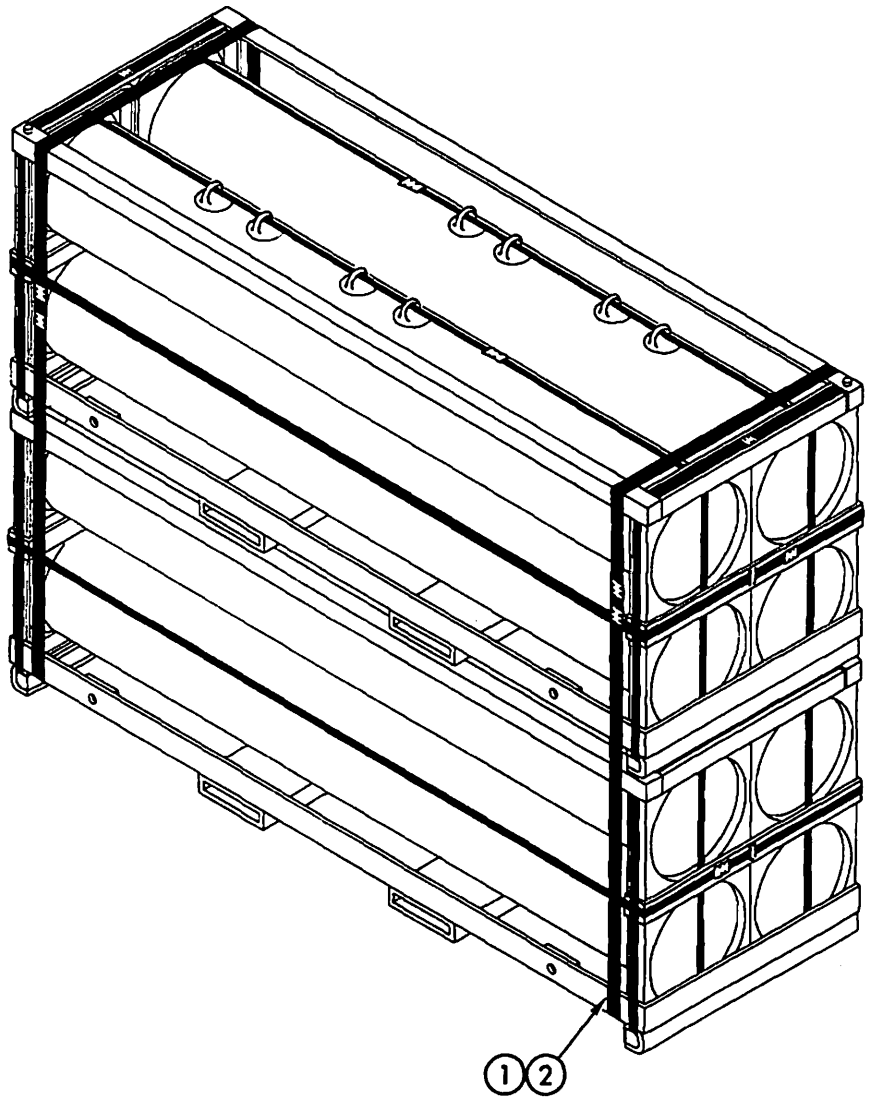
WHEN BOXCAR HAS STEEL DOOR POSTS DOORWAY PROTECTION AS PER MIL-STD-1325 (NAVY) PAGE 46, FIG. 21 IS REQUIRED.



DETAIL D
CENTER GATE
2 REQUIRED

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UNITIZING



1. STACK TWO UNIT LOADS USING FORK LIFT TRUCK OR SUITABLE HOISTING DEVICE EQUIPPED WITH A SUITABLE FOUR LEG BRIDLE.
2. BE CERTAIN THAT STACKING PINS OF THE BOTTOM UNIT LOAD PROPERLY ENGAGE THE UPPER UNIT LOAD.
3. SECURE TOP UNIT LOAD TO BOTTOM UNIT LOAD WITH TWO STEEL STRAPS (PIECE 1) POSITIONED AS INDICATED ABOVE.
4. SEAL WITH TWO SEALS (PIECE 2) EACH STRAP AND DOUBLE CRIMP.

REVIEW ACTIVITY:
NAVY - OS, AS

PREPARING ACTIVITY:
NAVY - OS
(PROJECT NO. 8140-N247)

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